

SHEBOYGAN COUNTY AP, WI, USA

WMO: 726425

Lat: 43.769N

Lon: 87.851W

Elev: 227

StdP: 98.62

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04841

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.6	-18.0	-25.8	0.4	-20.1	-23.2	0.5	-17.6	11.7	-6.4	10.7	-6.1	4.5	280	0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	31.2	23.3	29.1	21.9	27.7	21.1	24.3	29.1	23.3	27.7	22.3	26.3	5.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.9	27.4	21.9	17.0	26.2	21.0	16.1	25.1	74.5	29.1	70.2	27.7	66.4	26.3	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.8	9.2	8.3	-24.0	33.7	2.9	2.0	-26.1	35.1	-27.8	36.3	-29.4	37.5	-31.5	38.9		
			-24.3	26.1	2.8	1.1	-26.3	26.9	-27.9	27.5	-29.5	28.1	-31.5	28.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-6.6	-5.7	0.3	6.4	12.5	18.1	21.1	20.0	16.2	9.6	3.1	-3.6	(6)
	DBStd	10.85	6.36	6.06	5.86	4.48	4.44	3.90	3.37	3.04	4.17	4.59	5.07	5.96	(7)
	HDD10.0	2115	515	440	307	125	24	1	0	0	3	65	214	422	(8)
	HDD18.3	4133	773	673	561	357	189	52	12	17	88	273	458	680	(9)
	CDD10.0	1270	0	0	5	18	100	243	345	311	190	53	6	0	(10)
	CDD18.3	246	0	0	0	0	7	44	98	69	26	2	0	0	(11)
	CDH23.3	2112	0	0	1	12	84	401	873	517	202	22	0	0	(12)
	CDH26.7	575	0	0	0	1	17	107	277	125	44	4	0	0	(13)
(14) Wind	WSAvg	3.8	4.5	4.5	4.2	4.5	3.9	3.3	3.0	2.8	3.1	3.8	4.1	4.3	(14)
(15) Precipitation	PrecAvg	817	37	34	47	82	96	102	88	83	74	69	54	49	(15)
	PrecMax	1004	84	98	107	128	253	261	200	155	133	136	142	102	(16)
	PrecMin	626	11	3	8	29	24	32	33	25	14	18	10	17	(17)
	PrecStd	93	22	21	29	30	50	53	39	35	32	34	34	24	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	10.9	20.2	24.9	28.8	32.4	34.1	32.6	30.6	26.3	19.1	12.0	(19)
		MCWB	5.7	7.5	15.1	16.8	20.9	23.4	25.1	23.8	22.3	18.9	13.5	10.3	(20)
	2%	DB	5.4	6.8	14.4	20.7	25.9	29.4	31.7	29.5	27.7	22.4	15.6	7.9	(21)
		MCWB	3.3	4.4	11.2	13.8	18.7	21.5	23.9	22.6	21.0	17.1	11.3	5.7	(22)
	5%	DB	3.4	3.9	11.2	17.1	23.1	27.5	29.4	27.8	25.8	19.5	12.7	6.0	(23)
		MCWB	2.1	2.1	8.0	11.7	17.0	20.8	22.4	21.4	20.0	15.3	9.6	4.6	(24)
	10%	DB	2.1	2.3	8.0	14.0	21.0	25.8	27.8	26.2	23.6	17.3	10.6	4.0	(25)
		MCWB	1.0	0.9	5.4	9.0	15.6	19.4	21.5	20.4	18.8	14.0	8.2	2.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	7.8	15.7	17.7	21.7	24.7	26.3	25.5	23.7	21.1	14.3	10.7	(27)
		MCDB	8.1	10.3	19.7	22.6	27.6	30.7	32.0	30.8	28.4	25.0	18.2	11.2	(28)
	2%	WB	3.6	4.6	11.8	14.9	19.7	22.9	24.7	23.9	22.2	18.2	12.1	6.5	(29)
		MCDB	4.8	6.5	14.4	19.9	24.1	28.0	30.0	27.9	26.1	20.5	14.4	7.2	(30)
	5%	WB	2.3	2.4	8.0	12.2	18.2	21.7	23.6	22.6	21.1	16.2	10.2	4.6	(31)
		MCDB	3.4	3.9	10.4	16.1	22.2	26.2	28.1	26.3	24.3	18.9	12.5	5.9	(32)
	10%	WB	1.0	0.9	5.6	9.8	16.4	20.3	22.6	21.5	19.8	14.3	8.1	2.8	(33)
		MCDB	2.1	2.0	8.0	13.6	20.1	24.4	26.7	24.9	22.8	16.9	10.0	4.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	8.3	8.5	9.7	10.7	10.8	11.2	10.7	11.3	9.6	8.3	7.2	(35)
		MCDBR	8.0	9.0	12.3	14.9	14.0	12.9	12.5	12.3	13.3	13.4	12.0	8.3	(36)
	5% WB	MCWBR	6.8	7.2	8.9	9.4	8.4	7.1	6.4	6.4	7.4	8.6	8.7	7.1	(37)
		MCWBR	7.7	8.8	11.9	14.0	13.0	11.6	11.3	10.9	11.6	11.4	11.0	7.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.281	0.314	0.337	0.383	0.405	0.423	0.415	0.408	0.382	0.340	0.304	0.282	(40)
		taud	2.430	2.322	2.378	2.292	2.281	2.278	2.307	2.316	2.377	2.486	2.520	2.475	(41)
	Ebn at Noon		856	875	902	880	868	850	853	846	842	840	824	821	(42)
		Edh at Noon	73	97	106	125	130	131	126	121	105	83	68	64	(43)
(44) All-Sky Solar Radiation	RadAvg		1.49	2.46	3.63	4.54	5.39	5.99	6.14	5.25	4.20	2.66	1.66	1.17	(44)
	RadStd		0.20	0.27	0.36	0.43	0.45	0.42	0.27	0.34	0.34	0.29	0.15	0.12	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page